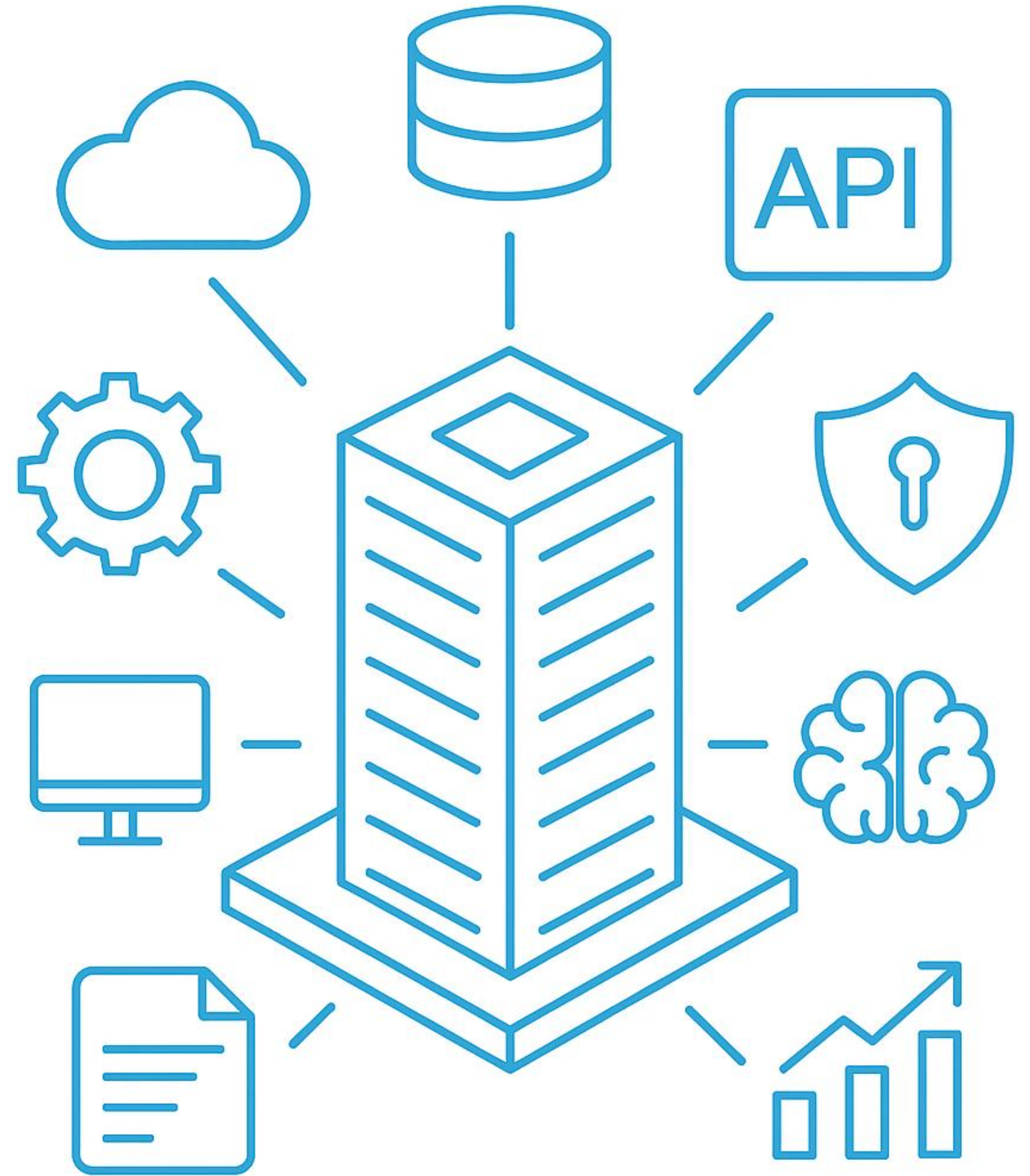
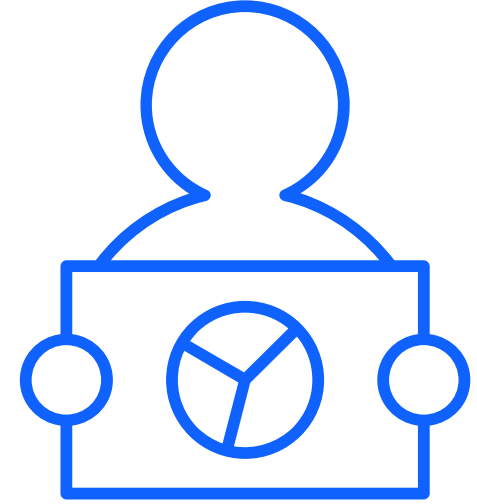
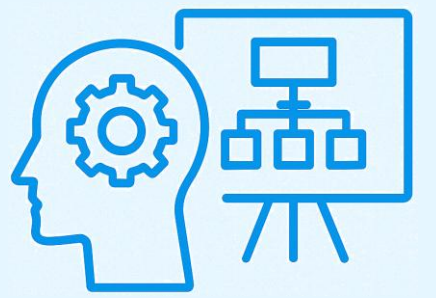


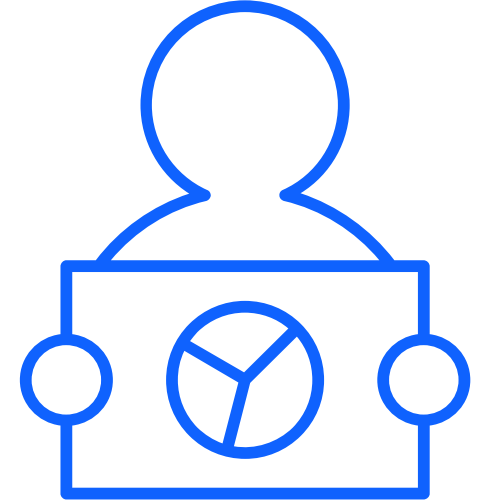
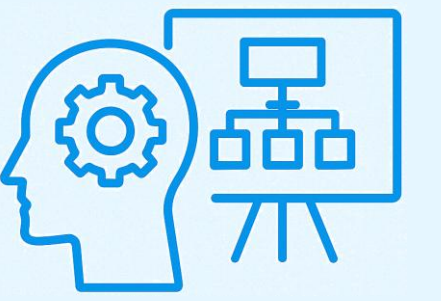
Architecture Overview





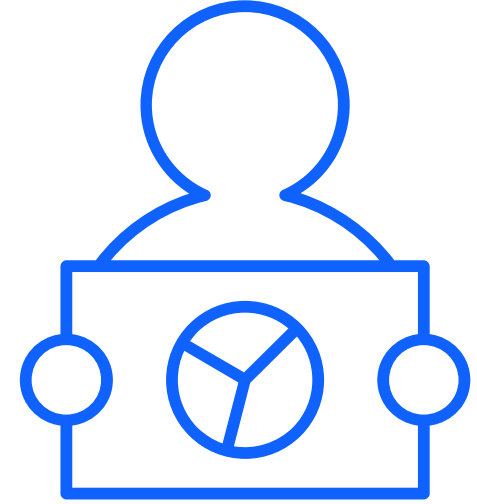
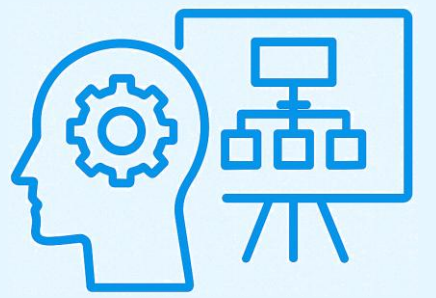
→ **Introduction and objectives**

- ↘ What is an Architecture Overview?
- ↘ Examples
- ↘ Best Practices
- ↘ Anti-patterns
- ↘ Summary and references

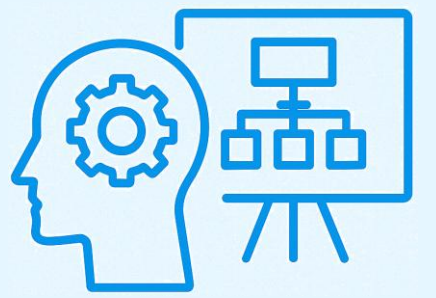


At the end of this module, you should be able to:

- Explain an Architecture Overview
- Describe the uses of an Architecture Overview
- Explain how an Architecture Overview fits in with the Component and Operational Models
- Develop a simple Architecture Overview
- Identify potential issues when reviewing an Architecture Overview



- ↗ Introduction and objectives
- **What is an Architecture Overview?**
- ↘ Examples
- ↘ Best Practices
- ↘ Anti-patterns
- ↘ Summary and references



From the Work Product Description

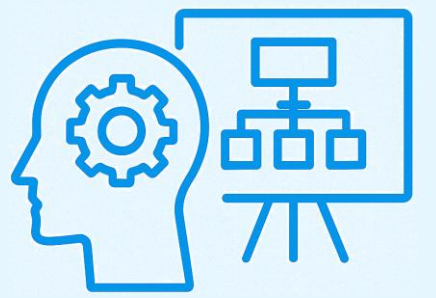
Purpose:

- To communicate to the sponsor and external stakeholders a conceptual understanding of the intended IT system
- To provide a high-level shared vision of the architecture and scope of the proposed IT system for the development teams
- To explore and evaluate alternative architectural options
- To enable early recognition and validation of the implications of the architectural approach
- To facilitate effective communication between different communities of stakeholders and developers
- To facilitate orientation for new people who join the project

Main Description:

This artifact provides an overview of the ‘main conceptual elements and relationships’ of an architecture, which may include candidate subsystems, components, nodes, connections, data stores, users, and external systems. As such, it represents the governing ideas and ‘candidate building blocks’ of the architecture.

What is an Architecture Overview (2 of 2)



From the Work Product Description

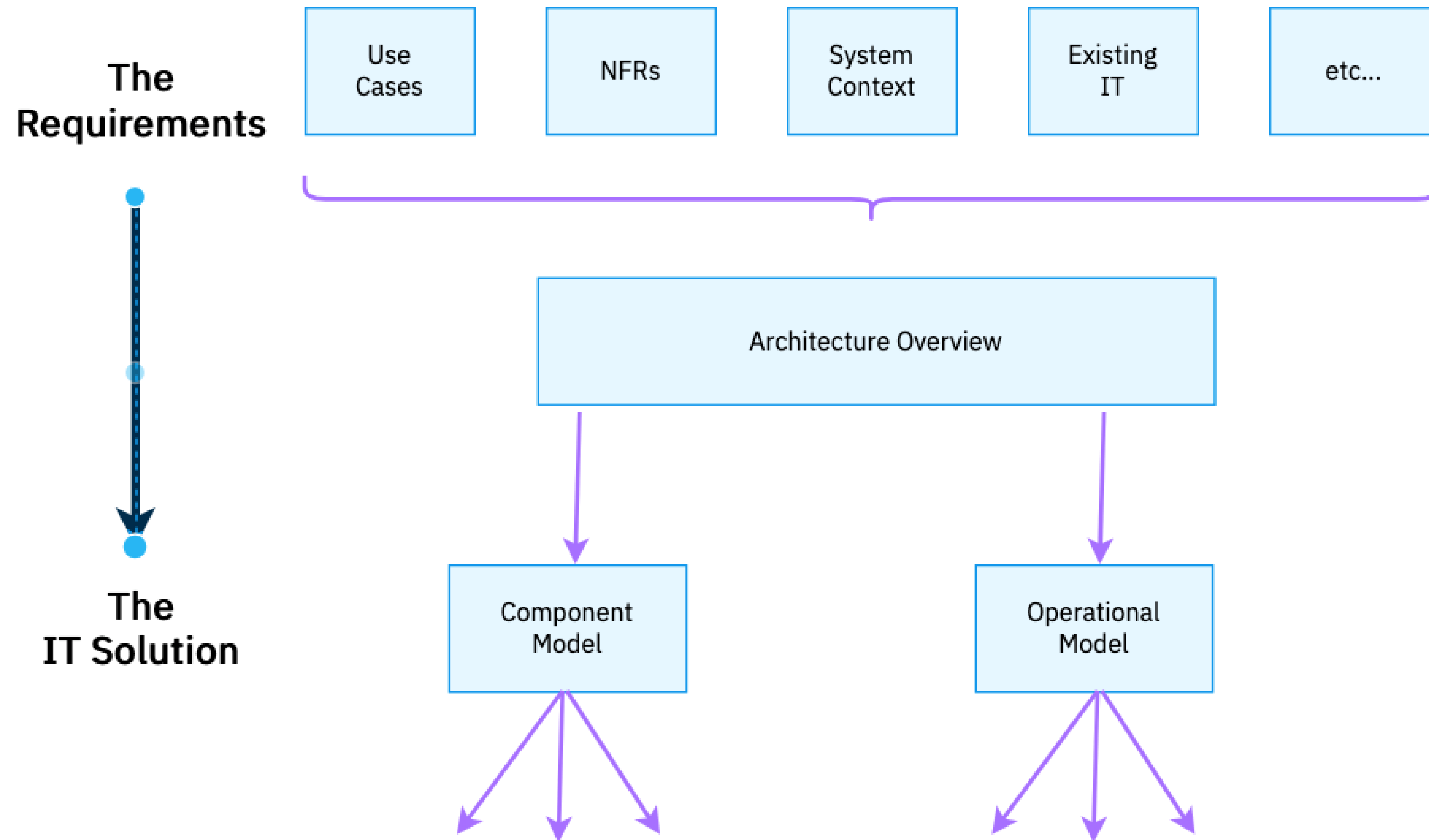
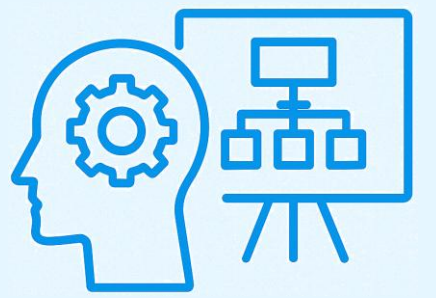
Notation:

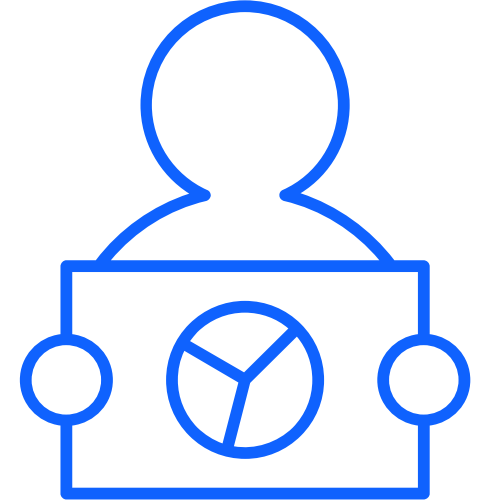
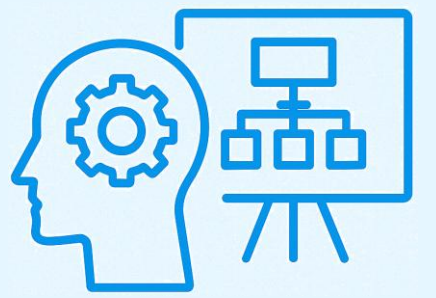
Consists of one or more ‘informal diagrams’ (in other words, not necessarily created using formal modeling tools or ‘notation’) together with supporting text describing the major architectural concepts of the architecture, such as:

- The range of different delivery mechanisms the architecture supports, such as home Internet terminals and kiosks
- The separation of functions
- The architecture model, such as a three-tier or four-tier model
- The use of hardware, such as distributed servers and workstations
- The access to legacy systems

- Diagrams are usually ‘static in nature’, showing the relationships between components.
- However, where it makes sense to do so, dynamic views showing collaborations between the architectural components may also be included.
- The diagrams may show ‘functional views, operational views, or a combination’ of the two.
- Where ‘alternative architectural solutions’ are being explored, diagrams may be produced for each option to enable various stakeholders to discuss the tradeoffs between the options.

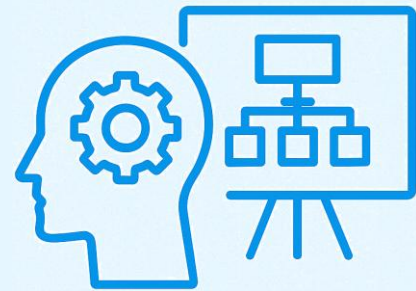
Where does the Architecture Overview fit?





- ↗ Introduction and objectives
- ↗ What is an Architecture Overview?
- **Examples**
- ↘ Best Practices
- ↘ Anti-patterns
- ↘ Summary and references

Example 1: Retail Industry – Customer Insights Architecture Overview



Business Roles



Line-of-Business Executive



Chief Marketing Officer



Chief Data Officer



Customer Officer



Agent

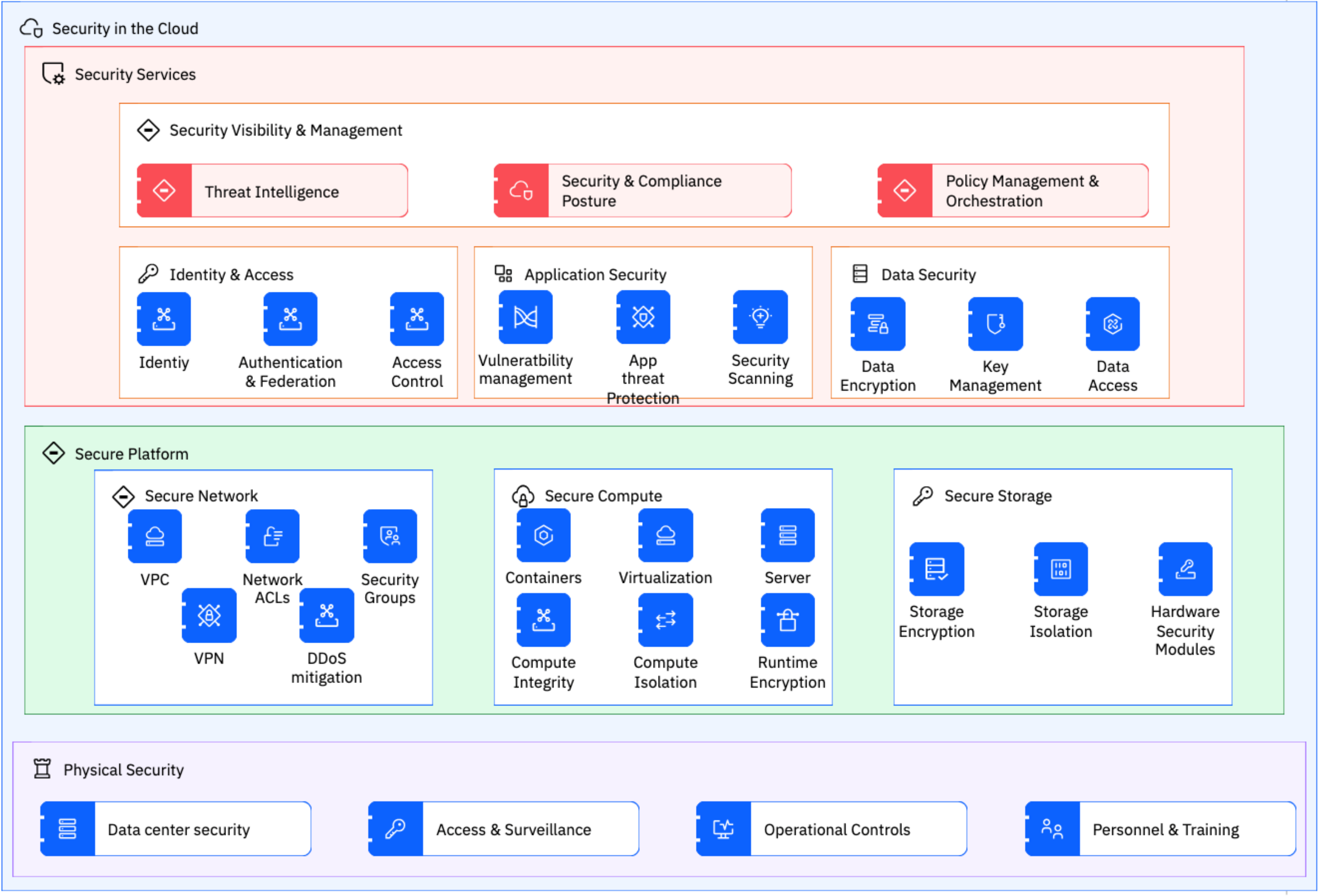
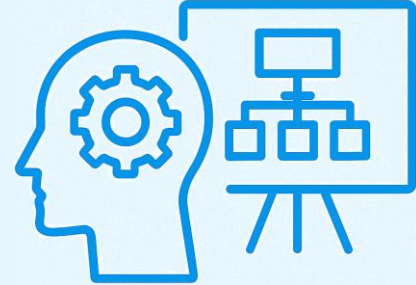
Pre-Integrated Capabilities

Deliver Insights 	Insights via API	UI Design & Pre-Built Dashboards
Perform Analytics 	Pre-Built Industry Analytic Models	Descriptive to Predictive to Cognitive
Prepare Data 	Industry Data Models & Connectors	Internal & External Data

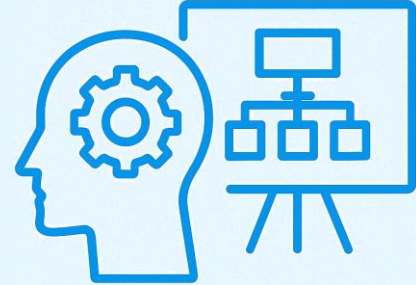
Relevant Data Sources



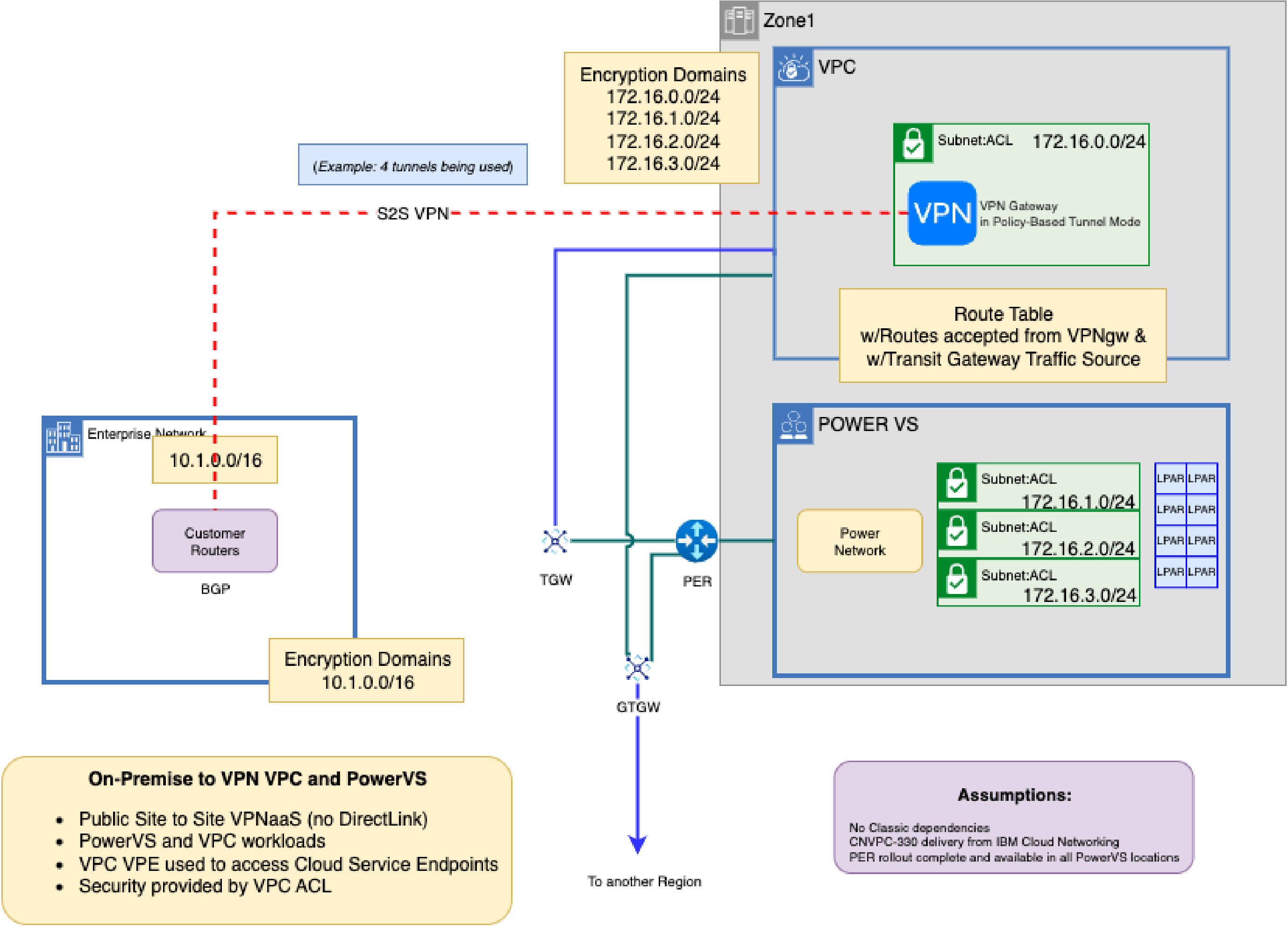
Example 2: Security in the Cloud



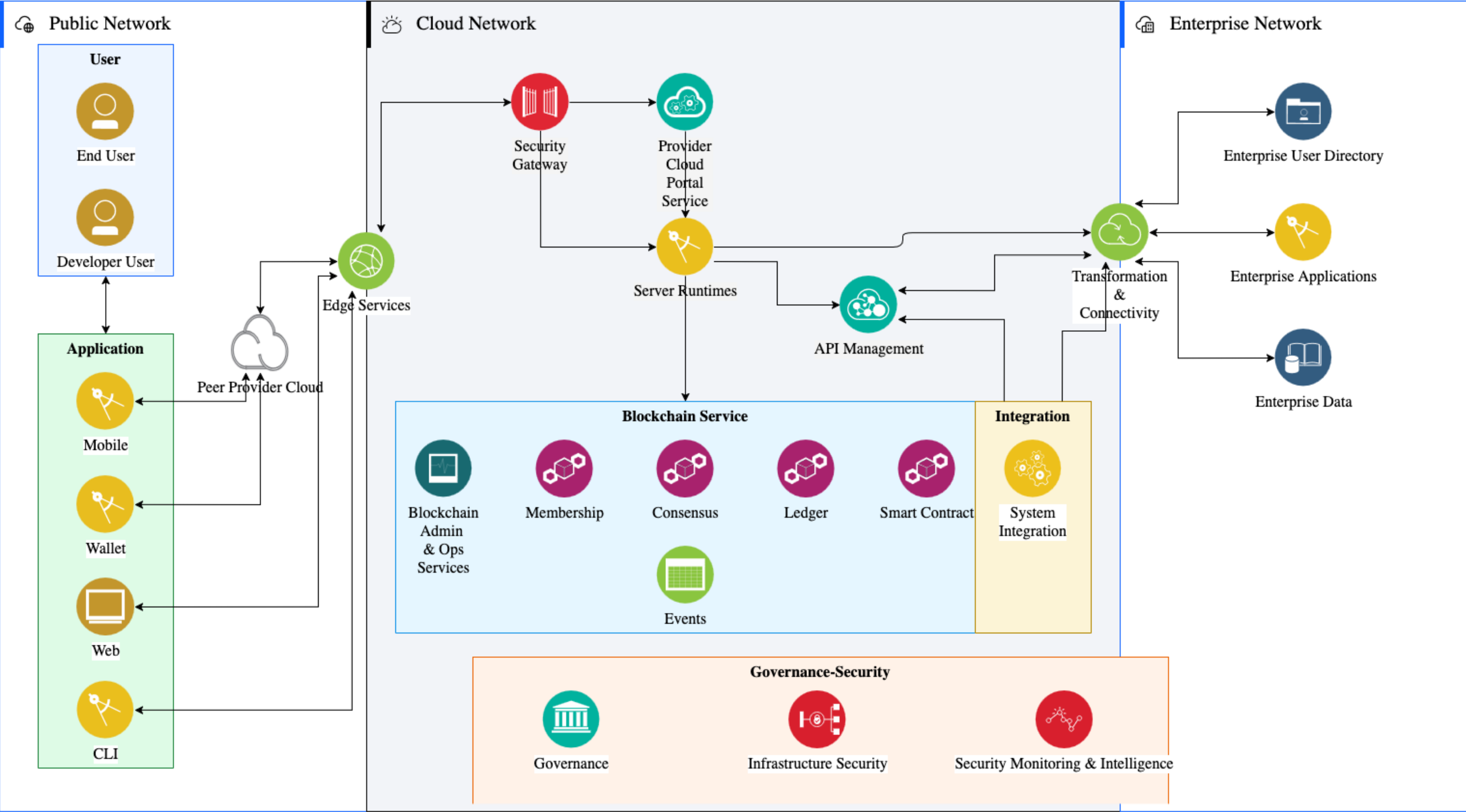
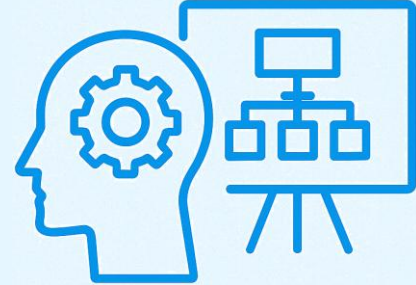
Example 3: Site to Site VPC VPN and PowerVS

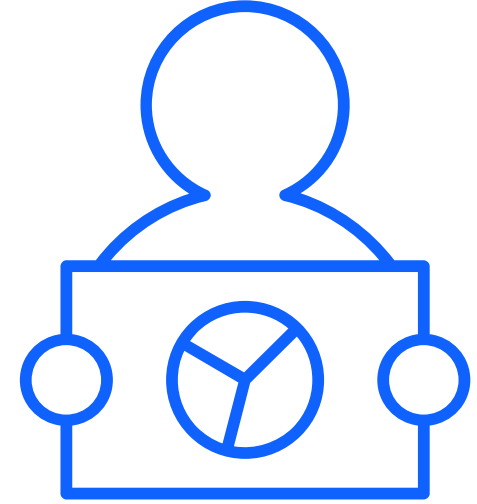
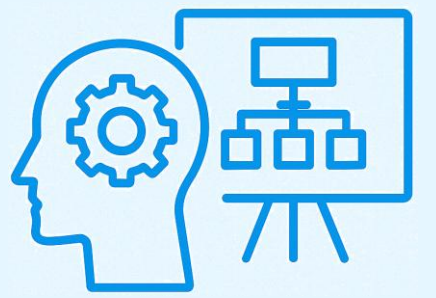


Site to Site VPC VPN and PowerVS



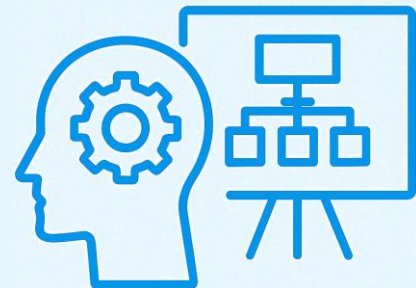
Example 4: Blockchain Application



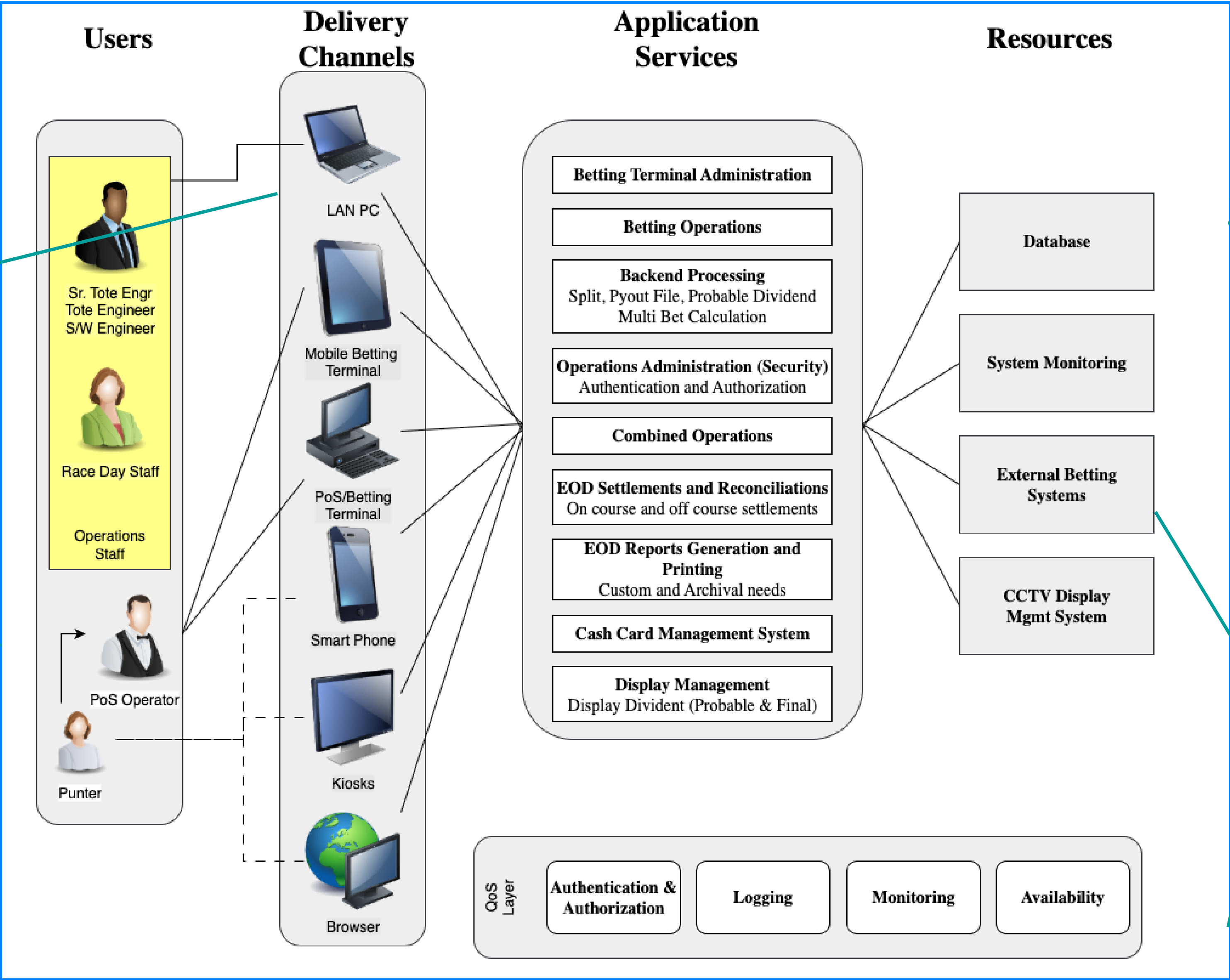


- ↗ Introduction and objectives
- ↗ What is an Architecture Overview?
- ↗ Examples
- **Best Practices**
- ↘ Anti-patterns
- ↘ Summary and references

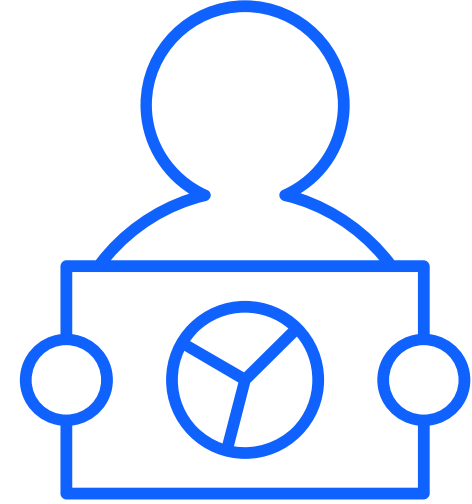
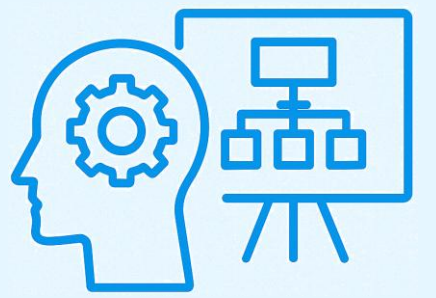
Best practices



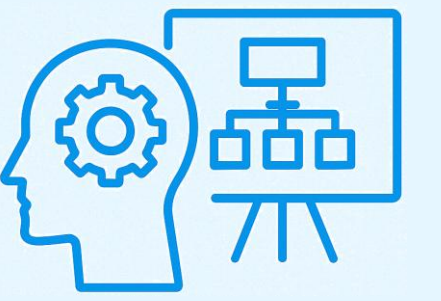
- #1 AO must clearly support the value proposition conversation of the proposed solution.
- #2 Ensure the Architecture View supports the target stakeholder profile. For example, Enterprise View for Business Users.
- #3 Do not stop at Enterprise View! Each AO View promotes a different yet related “Architecture Thinking”.
- #4 Leverage Reference Architectures to expedite development of an AO.
- #5 In some AO instances “Less may be more”. Avoid over crowding an AO.



- #6 Clearly depict the architecture building blocks of the proposed solution.
- #7 Depict Architecture Thinking towards Non-Functional requirements.
- #8 Depict higher level abstractions rather than fine grained details.
- #9 Leverage existing System names (for example, External Betting System) rather than generic (such as Legacy system).
- #10 Use Annotations to convey more. For example, Scope of release, systems management, high availability, and so on.



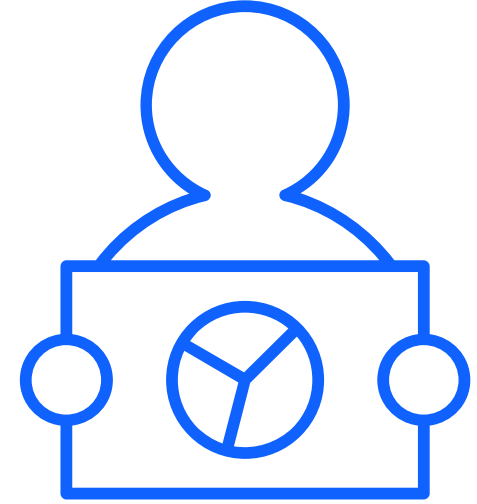
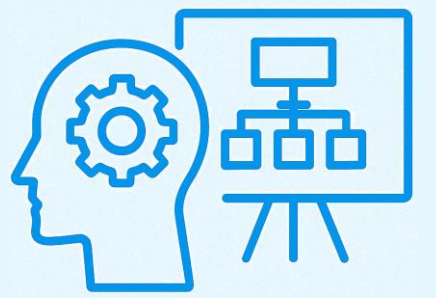
- ↗ Introduction and objectives
- ↗ What is an Architecture Overview?
- ↗ Examples
- ↗ Best Practices
- **Anti-patterns**
- ↘ Summary and references



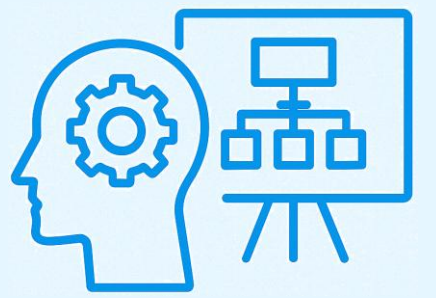
Brainstorming:

What do you consider to be anti-patterns?

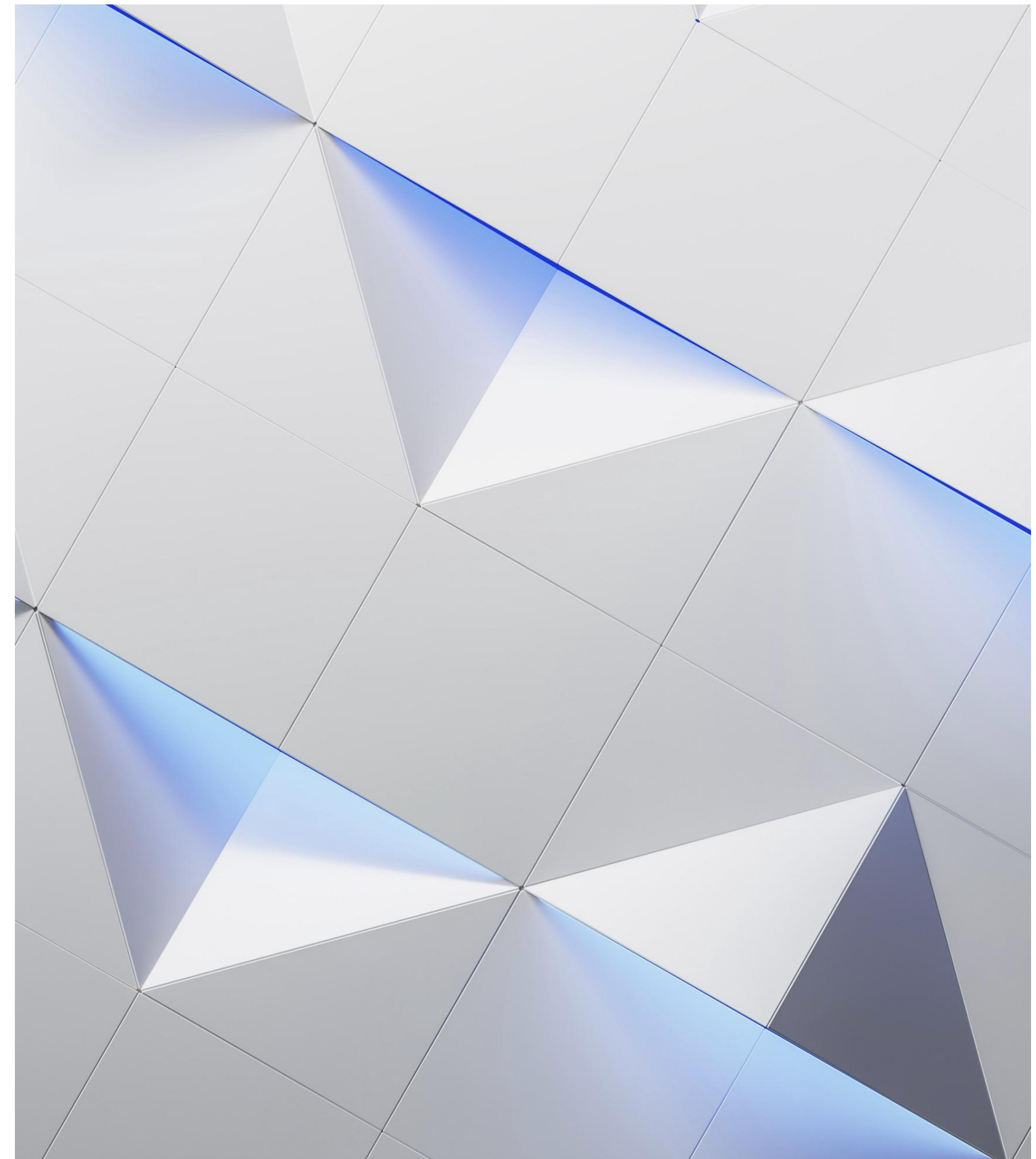
- Information overload: Putting too much information into one diagram
- Information underload: Putting too little information into the Architecture Overview
- Using too many acronyms that are not properly introduced
- Providing more technical details than required for a business audience

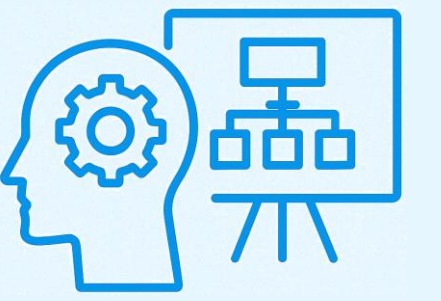


- ↗ Introduction and objectives
- ↗ What is an Architecture Overview?
- ↗ Examples
- ↗ Best Practices
- ↗ Anti-patterns
- **Summary and references**



- The main purpose of an Architecture Overview is communication.
- It is more important for the Architecture Overview to be simple, brief, clear, and understandable than comprehensive or accurate in all details.
- The Architecture Overview represents the governing ideas and candidate building blocks of an IT system or enterprise architecture.
- The Architecture Overview provides an overview of the main conceptual elements and relationships in an architecture.





Architecture Overview, Work Product ART 512, on MethodWorkspace

Thank you

IBM, the IBM logo, and ibm.com are trademarks or registered trademarks of International Business Machines Corporation in the United States, other countries, or both. If these and other IBM trademarked terms are marked on their first occurrence in this information with a trademark symbol (® or TM), these symbols indicate U.S. registered or common law trademarks owned by IBM at the time this information was published. Such trademarks may also be registered or common law trademarks in other countries. A current list of IBM trademarks is available on the Web at "Copyright and trademark information." Java and all Java-based trademarks and logos are trademarks or registered trademarks of Oracle and its affiliates.

Linux is a trademark of Linus Torvalds in the United States, or other countries, or both. Microsoft, Excel, PowerPoint, Project, Visio, Word, and Windows are trademarks of the Microsoft Corporation in the United States, other countries, or both. UML is a registered trademark of Object Management Group, Inc. in the United States and/or other countries. TOGAF and UNIX are registered trademarks of The Open Group in the United States and other countries. Other company, product, and service names may be trademarks of IBM or other companies. Citrix is a trademark of Citrix Systems, Inc. and one or more of its subsidiaries, and may be registered in the United States Patent and Trademark Office and in other countries.

